

Scientific Computing Exercise Set 2

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Repository: (*Scientific Computing*) https://github.com/ShioMura/Scientific_Computing

I. INTRODUCTION

Diffusion-driven processes are fundamental in physics, chemistry, and materials science. In the steady-state regime and in the absence of internal sources, diffusion is governed by Laplace's equation

$$\nabla^2 c = 0, \quad (1)$$

which describes harmonic scalar fields. When such a diffusive field is coupled to a moving boundary, complex growth patterns may emerge due to nonlinear feedback between geometry and flux.

The aim of this project is to implement a numerical solver for the two-dimensional Laplace equation using the Successive Over-Relaxation (SOR) method, to validate the solver under prescribed boundary conditions including an internal absorbing region, and to simulate diffusion-limited aggregation (DLA) using harmonic measure. Furthermore, the influence of a nonlinear growth parameter η on the morphology of the aggregate is investigated. This study demonstrates how deterministic diffusion governed by Laplace's equation, when coupled to stochastic boundary evolution, gives rise to nonlinear Laplacian instabilities and fractal growth.

II. THEORY

A. Laplace Equation

In two spatial dimensions, Laplace's equation reads

$$\nabla^2 c = \frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} = 0. \quad (2)$$

Solutions to this equation are harmonic functions and satisfy the mean value property, meaning that the value at any interior point equals the average over its surrounding neighborhood. In diffusion problems, this corresponds to steady-state concentration fields.

B. Finite Difference Discretization

To obtain a numerical solution, the continuous domain is discretized onto a uniform square grid. Using the standard five-point stencil, the Laplace equation is approximated by

$$c_{i,j} = \frac{1}{4} (c_{i+1,j} + c_{i-1,j} + c_{i,j+1} + c_{i,j-1}), \quad (3)$$

which transforms the partial differential equation into a linear algebraic system defined at each grid point.

C. Successive Over-Relaxation

The resulting linear system is solved iteratively using the Successive Over-Relaxation (SOR) method. The update rule is given by

$$c_{i,j}^{\text{new}} = (1-\omega)c_{i,j}^{\text{old}} + \frac{\omega}{4} (c_{i+1,j} + c_{i-1,j} + c_{i,j+1} + c_{i,j-1}), \quad (4)$$

where $\omega \in (1, 2)$ is the relaxation parameter. Values $\omega > 1$ accelerate convergence relative to standard Gauss-Seidel iteration. The iteration is continued until convergence is achieved, defined by

$$\max_{i,j} |c_{i,j}^{\text{new}} - c_{i,j}^{\text{old}}| < \text{tol}. \quad (5)$$

D. Harmonic Measure and DLA

In diffusion-limited aggregation, growth occurs preferentially at boundary sites where the diffusive flux is highest. Since flux is proportional to the gradient of the concentration field, the probability of growth at a boundary site (i, j) is taken to be

$$p(i, j) \propto |\nabla c(i, j)|^\eta. \quad (6)$$

The parameter η controls the degree of nonlinear amplification. For $\eta = 1$, growth follows the harmonic measure. Larger values of η enhance tip instability and lead to thinner, more branched structures, while smaller values promote more compact growth. This coupling between diffusion and evolving geometry gives rise to screening effects and fractal morphologies characteristic of Laplacian growth.

E. Monte Carlo Diffusion-Limited Aggregation

In addition to harmonic-measure growth, diffusion-limited aggregation can be modeled using stochastic random-walker dynamics. In the Monte Carlo formulation, particles are released one at a time from the boundary of the computational domain and perform unbiased random walks until they reach the existing cluster.

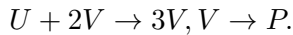
At each time step, a particle moves to one of its four nearest neighbors with equal probability. If the particle reaches a site adjacent to the aggregate, it sticks with probability p_s . The sticking probability introduces stochastic growth dynamics and allows investigation of screening effects without explicitly solving Laplace's equation.

For $p_s = 1$, particles stick immediately upon contact, leading to strongly tip-dominated growth. For $p_s < 1$, particles continue diffusing before attachment, reducing tip amplification and producing more compact aggregates.

This Monte Carlo approach provides a direct stochastic realization of diffusion-limited aggregation and serves as a reference model for comparison with harmonic-measure growth.

F. Gray-Scott model

The Gray-Scott model is a reaction-diffusion system that describes how two reactions influence the concentrations of the relevant chemicals. In this report we will discuss the following two reactions



The model describing these concentrations consist of two nonlinear partial differential equations, including parameters describing diffusion rates and the supply and decay of the chemicals.

III. METHODS

All simulations were performed on a uniform square grid of size $N = 100$. The Laplace equation was solved iteratively using the Successive Over-Relaxation (SOR) method with relaxation parameter $\omega = 1.8$. Convergence was achieved when the maximum absolute update between successive iterations satisfied $\max_{i,j} |c_{i,j}^{\text{new}} - c_{i,j}^{\text{old}}| < 10^{-6}$.

Dirichlet boundary conditions were imposed by fixing the concentration to $c = 1$ along the top boundary and $c = 0$ along the remaining boundaries. To validate the solver, an internal square region at the center of the domain was fixed to zero concentration, representing an

absorbing object. The harmonic field was recomputed until convergence for each configuration.

For the diffusion-limited aggregation simulations, a single seed particle was initially placed at the center of the domain. At each growth step, the Laplace equation was solved for the current geometry. Empty lattice sites adjacent to the aggregate were identified, and the gradient magnitude of the concentration field was evaluated at those sites using central finite differences. Growth probabilities were assigned according to $p(i, j) \propto |\nabla c(i, j)|^\eta$, and one site was selected randomly using normalized probabilities. The selected site was then added to the aggregate.

For each value $\eta \in \{0.5, 1.0, 2.0\}$, 300 growth steps were performed. All simulations were implemented in Python using NumPy for numerical computations and Matplotlib for visualization.

A. Monte Carlo DLA Implementation

For the Monte Carlo DLA simulations, a single seed particle was placed at the center of the domain. Particles were released sequentially from random positions near the boundary and allowed to perform unbiased random walks.

At each step, the particle moved randomly to one of its four neighboring lattice sites. If the particle reached a site adjacent to the cluster, it stuck with probability p_s . If the sticking condition was not satisfied, the random walk continued until attachment occurred.

Simulations were performed for $p_s \in \{0.2, 0.5, 1.0\}$ using 200 particles per configuration. This allowed comparison between strongly adhesive growth and more diffusive aggregation regimes.

All simulations were implemented in Python using NumPy for random number generation and array manipulation.

B. Gray-Scott model

The Gray-Scott is implemented using the Neumann boundary conditions, which means that both $\frac{\partial u}{\partial n}$ and $\frac{\partial v}{\partial n}$ are taken as 0. This choice is motivated by the fact that in the real world chemicals can not leave or enter true the boundaries. In our implementation this is enforced by setting the boundaries equal to the interior point. The Gray-Scott model is implemented by dividing the grid into cells of size $\Delta x \times \Delta y$.

After which for each timestep we estimate $\nabla^2 u$ and $\nabla^2 v$ with the five point stencil.

$$\nabla^2 u = \frac{u_{i+1,j} + u_{i,j-1} + u_{i,j+1} + u_{i-1,j} - 4u_{i,j}}{\Delta x^2},$$

and in a similar way for v .

IV. RESULTS

A. Laplace Solver Verification

The SOR solver was first tested on a square domain with fixed boundary conditions. The top boundary was set to $c = 1$, while the remaining boundaries were fixed at $c = 0$. The numerical solution converged rapidly for relaxation parameters $\omega \in [1.5, 1.9]$, with the maximum update falling below the prescribed tolerance within a few hundred iterations.

The resulting concentration field exhibits the expected harmonic behavior, with smooth monotonic decay from the top boundary toward the bottom. When an internal square sink was introduced by fixing a central region to zero concentration, the equipotential lines bend around the absorbing region, confirming correct implementation of internal Dirichlet boundary conditions.

B. Harmonic-Measure DLA Growth

Starting from a single seed particle at the center of the domain, the aggregate was grown by iteratively solving Laplace's equation and selecting boundary sites according to the harmonic measure.

For $\eta = 1$, the cluster develops a branched fractal structure. Growth occurs predominantly at exposed tips, while interior regions become screened from the diffusive flux. This behavior is characteristic of classical DLA and demonstrates correct coupling between the Laplace field and boundary evolution.

C. Influence of the Nonlinear Parameter η

To investigate the effect of nonlinear growth amplification, simulations were performed for $\eta = 0.5$, $\eta = 1.0$, and $\eta = 2.0$ (see Figure 1). For $\eta = 0.5$, growth is more uniform and the resulting cluster is relatively compact. The reduced nonlinear amplification weakens tip selection and allows growth to occur more evenly along the boundary.

For $\eta = 1.0$, the aggregate exhibits the classical fractal morphology associated with harmonic measure, featuring multiple branching arms and clear screening effects.

For $\eta = 2.0$, the growth becomes strongly tip-dominated. The aggregate develops thin, needle-like structures with pronounced instability. Screening is enhanced, and growth concentrates almost exclusively at the most exposed regions.

These results confirm the expected transition from compact to highly unstable Laplacian growth as η increases.

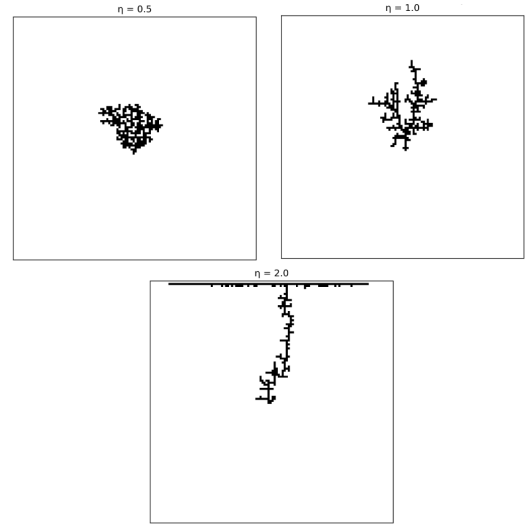


Fig. 1: Comparison of DLA clusters for $\eta = 0.5$, $\eta = 1.0$, and $\eta = 2.0$. Increasing η enhances tip amplification and leads to thinner, more unstable growth.

D. Monte Carlo DLA

Monte Carlo simulations reproduce the characteristic fractal morphology of diffusion-limited aggregation. For $p_s = 1.0$, particles stick immediately upon contact, leading to highly branched and tip-dominated structures.

For lower sticking probabilities, particles diffuse longer before attachment. This reduces screening and results in comparatively more compact clusters. In particular, for $p_s = 0.2$, the aggregate appears less sharply branched than in the fully adhesive case.

These results confirm that stochastic diffusion alone is sufficient to generate fractal growth patterns, even without explicitly solving Laplace's equation.

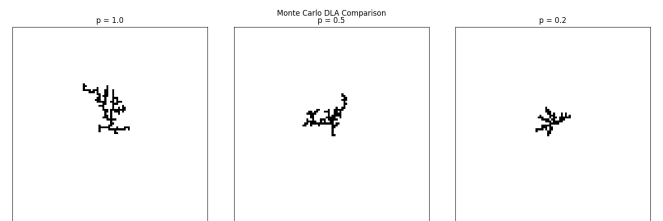


Fig. 2: Comparison of Monte Carlo DLA clusters for different sticking probabilities p_s . Lower p_s values reduce tip amplification and produce more compact aggregates.

E. Gray-Scott model

To investigate the resulting concentration of both V and U over time we run the model with its initial conditions. We now focus on the impact of the feeding rate f and removal rate r .

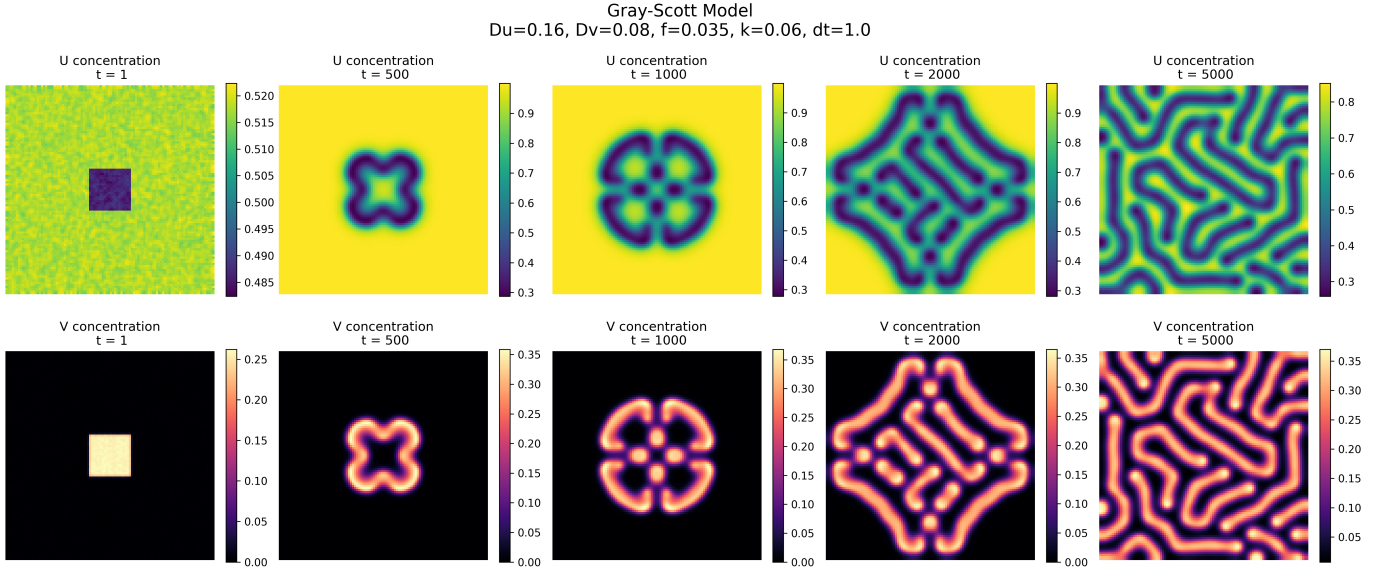


Fig. 3: Time evolution of the Gray–Scott reaction–diffusion model. The concentrations of U (top row) and V (bottom row) are shown at times $t = 1, 500, 1000, 2000, 5000$. Starting from a small square of V in a filled grid of U , with small local perturbation. The interaction between these two chemicals lead to the emergence of complex spatial patterns.

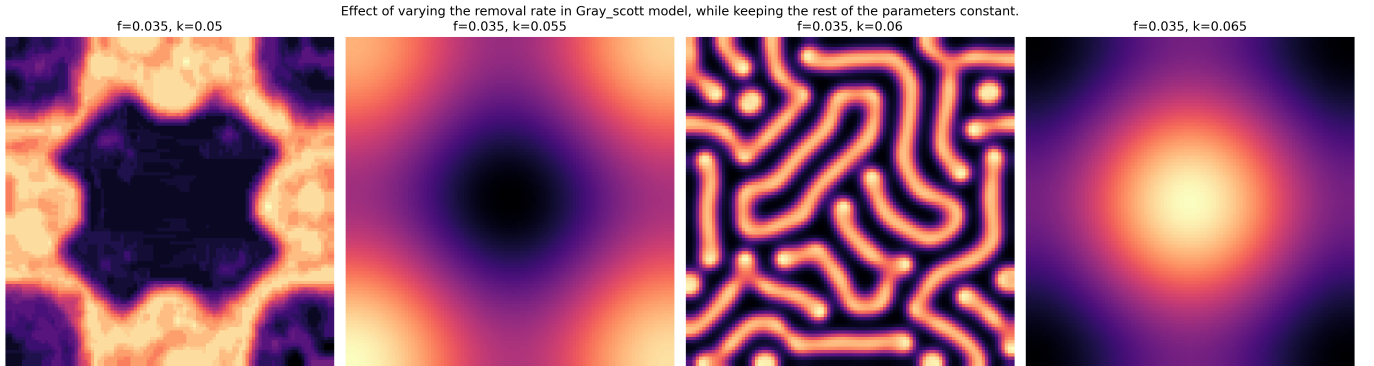


Fig. 4: Effect of varying the removal rate k in the Gray–Scott model, while keeping $f = 0.035$ fixed. Changes in k alter the decay rate of chemical V , which significantly affects the stability and morphology of the resulting patterns.

V. DISCUSSION

The numerical results confirm that the SOR method provides a stable and efficient solver for the Laplace equation under Dirichlet boundary conditions. The introduction of an internal absorbing region demonstrates correct handling of fixed interior constraints and validates the implementation. Convergence was achieved reliably for relaxation parameters in the range $\omega \in [1.5, 1.9]$, confirming the effectiveness of over-relaxation in accelerating the iterative scheme.

The harmonic-measure DLA simulations reproduce the characteristic features of Laplacian growth, including

tip amplification and screening. Growth occurs preferentially at exposed boundary regions where the diffusive flux is maximal, while interior regions become progressively shielded. This screening mechanism is a fundamental property of diffusion-limited aggregation.

The parameter η controls the nonlinear amplification of growth probability. For $\eta < 1$, growth becomes more uniform and compact. For $\eta = 1$, classical fractal branching emerges. For $\eta > 1$, tip selection is enhanced, leading to highly unstable, needle-like morphologies. This transition illustrates the mechanism underlying Laplacian instability, where small protrusions are amplified by the diffusive field.

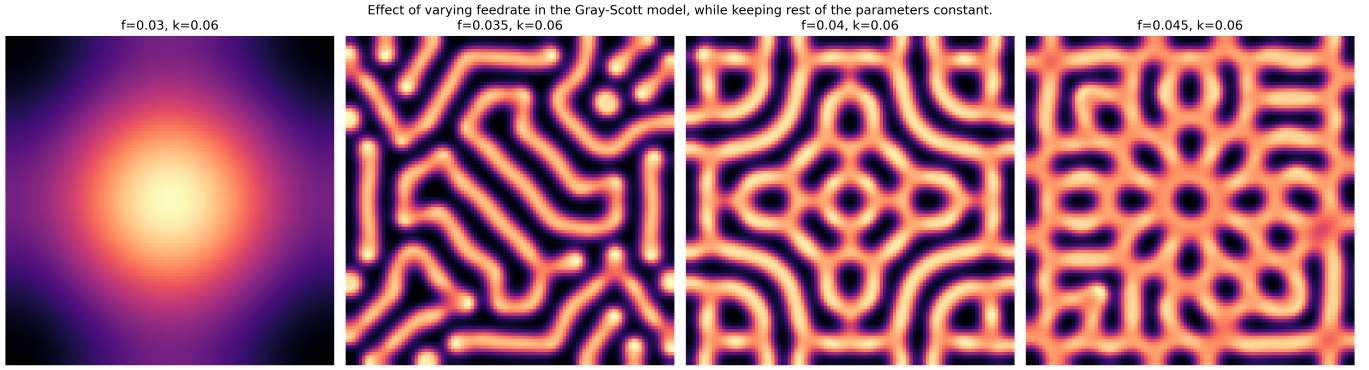


Fig. 5: Effect of varying the feed rate f in the Gray–Scott model, while keeping $k = 0.06$ fixed. Increasing f changes the balance between feeding chemical U and V , leading to different spatial structures ranging from uniform states to complex labyrinth-like patterns.

The Monte Carlo DLA simulations provide a complementary stochastic realization of diffusion-driven growth. While the harmonic-measure approach computes growth probabilities from the solution of Laplace’s equation, the Monte Carlo method reproduces similar screening and branching behavior through explicit random walks. For large sticking probability p_s , growth is strongly tip-dominated. For smaller p_s , particles diffuse longer before attachment, reducing tip amplification and producing slightly more compact aggregates. The qualitative agreement between the two approaches confirms the consistency between stochastic diffusion and harmonic-measure-driven growth.

Limitations of the model include finite grid effects, possible directional bias introduced by boundary conditions, and computational cost due to repeated Laplace solves in the harmonic formulation. Despite these limitations, the simulations successfully capture the essential physics of diffusion-limited aggregation.

A. Gray–Scott Model

Figure 3 demonstrates the ability of the Gray–Scott reaction–diffusion model to generate complex spatial patterns from a nearly homogeneous initial condition with small perturbations. As the system evolves, diffusion and nonlinear reactions interact to produce labyrinth-like and spot-like structures.

Small variations in the feed rate f and removal rate k lead to qualitatively different long-term morphologies. For certain parameter values, localized spot-like structures emerge, whereas intermediate values generate complex labyrinthine patterns. The results highlight the strong sensitivity of reaction–diffusion systems to parameter changes, as illustrated in Figures 5 and 4.

VI. CONCLUSION

In this work, a two-dimensional Laplace solver based on the SOR method was implemented and validated. The solver was coupled to both a harmonic-measure growth algorithm and a stochastic Monte Carlo formulation to simulate diffusion-limited aggregation.

The simulations demonstrate how nonlinear amplification of diffusive flux leads to fractal branching and Laplacian instability. By varying the growth parameter η , a transition from compact to highly unstable morphologies was observed. The Monte Carlo simulations further confirmed that stochastic diffusion alone reproduces the essential screening and branching behavior of DLA.

In addition, the Gray–Scott reaction–diffusion model was implemented and shown to produce complex spatial patterns highly sensitive to parameter variations. Together, these results illustrate how diffusion coupled with nonlinear dynamics generates rich pattern formation phenomena.

This report went over the Gray–Scott reaction–diffusion model, implemented and simulated on a two-dimensional grid using a finite difference method. Starting from a small perturbation, complex spatial patterns emerged due to the interaction between diffusion and nonlinear reactions. It was observed that small changes in parameter values significantly affect the resulting patterns. These results illustrate how simple reaction–diffusion systems can produce rich and diverse pattern formation.